A Study on Enhancing Hydrogen Release Property in Calcium Hydride through Ball Milling

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Abstract: This study explores the effect of ball milling on calcium hydride (CaH_2) as a hydrogen storage medium. With hydrogen emerging as a clean energy alternative, metal hydrides like CaH_2 are promising due to their high storage capacity. However, their decomposition/formation efficiency is limited. Ball milling, known to alter material properties, may enhance CaH_2's performance. This work investigates how milling conditions affect its hydrogen release behavior

Keywords: Ball milling, Hydrogen Storage, Calcium Hydride

1 Introduction

The transition from fossil fuels to renewable energy sources is imperative to mitigate environmental degradation and climate change[1, 2, 3]. Hydrogen, with its high energy content and clean combustion, stands out as a leading alternative. Among the various hydrogen storage methods, metal hydrides offer a compact and safe solution[4, 5]. This research focuses on calcium hydride, a material known for its high hydrogen storage capacity but limited by kinetic and thermodynamic barriers. Previous studies have extensively explored composite hydrides but rarely delved into pure calcium hydride. We hypothesize that ball milling could significantly improve the hydrogen release performance of calcium hydride by altering its surface properties and particle size, potentially making it a more efficient storage medium.

2 Related Works

Numerous studies demonstrate that ball milling enhances the hydrogen storage properties of metal hydrides. Wu et al.[6] investigated $Mg_{17}Al_{12}$ combined with CaH_2 and achieved a high hydrogen generation yield of 94.8% under optimized hydrolysis conditions. Tessier et al.[7] explored MgH_2 mixed with Ca and CaH_2, showing that ball milling significantly accelerated hydrolysis, and achieved an 80% yield within 30 minutes

Schiavo et al.[8] studied the CaH_2 + MgB_2 system with AlB2 additives and found that ball milling improved reaction kinetics and hydrogen sorption characteristics, emphasizing the importance of both the milling process and additive selection. Leardini et al.[9] synthesized calcium amidoborane through reactive milling, revealing complex hydrogen release behaviors that further highlight how structural modifications influence storage performance.

Yuan et al. [10] conducted a detailed investigation on TiH_2 samples subjected to various ball-milling durations (10 min to 4 h). Their study demonstrated that ball milling significantly lowered the first-stage H_2 release temperature by approximately 150°C and improved hydrogen release kinetics by reducing crystallite and particle sizes and disrupting surface oxide layers. This work shows that optimized milling can improve hydrogen release, but prolonged milling may be detrimental due to compositional changes.

While these findings affirm the effectiveness of ball milling in improving hydrogen storage, most focus on composite systems. In contrast, this study targets pure calcium hydride to fill the current research gap and evaluate its standalone performance.

3 Experimental

3.1 Materials

Calcium hydride (CaH_2) powder (Sigma Aldrich, Percentage 95%) was obtained commercially and used without further purification. All procedures were carried out in an argon-filled glovebox.

3.2 Preparation of Ball-Milled CaH_2

The as-received CaH_2 powder was mechanically milled using a planetary ball mill (Fritsch PL-7, Fritsch). A total of 150 mg of commercial CaH_2 powder and three zirconium oxide (ZrO_2) balls (each 1 cm in diameter, 2.98 g) were placed into a 20 cm^3 milling container, giving a ball-to-powder weight ratio of approximately 150:1.

Milling was performed at 400 rpm for various durations (10 min, 30 min, 1 h, 2 h, and 4 h). To minimize temperature rise during prolonged milling, each cycle consisted of 5 minutes of rotation followed by a 30-second rest. All milling and handling of powders were done inside an argon-filled glovebox.

Additionally, a sample of 2h ball-milled CaH_2 was left to oxidize overnight to investigate the effects of air exposure on its properties.

3.3 Characterization

X-ray diffraction (XRD) measurements were conducted at room temperature (~25°C) using a benchtop diffractometer (Rigaku MiniFlex) with Cu Kα radiation. X-rays were generated using the line focus principle. A reflection-free silicon plate served as the sample holder, and a Kapton film capsule was used to prevent atmospheric exposure. The average crystallite size (D) was estimated using the Scherrer equation:

$$D = \frac{k\lambda}{\beta \cos\theta}$$

where k is a dimensionless shape factor (typically 0.89–0.9 for spherical crystals with cubic unit cells), λ is the X-ray wavelength, β is the full width at half maximum (FWHM) in radians (on the 2θ scale), and θ is the Bragg angle.

Scanning electron microscopy (SEM) was performed using a TM4000II/TM4000PlusII device at an accelerating voltage of 15kV. Powder samples were mounted on graphite tape attached to an aluminum stub.

X-ray photoelectron spectroscopy (XPS) was carried out using a JPS 9010 TR spectrometer equipped with an ultra-high vacuum chamber and a Mg Kα X-ray source (λ = 1253.6 eV). The pass energy was set to 10 eV. The samples were fixed on graphite tape, and the Shirley background was subtracted using *SpecSurf* software.

Thermogravimetric and differential thermal analysis (TG-DTA) was conducted in conjunction with hydrogen temperature-programmed desorption (H_2-TPD) using a STA-2500 Regulus analyzer under argon flow. The exhaust gases were analyzed with a mass spectrometer to detect H_2 production.

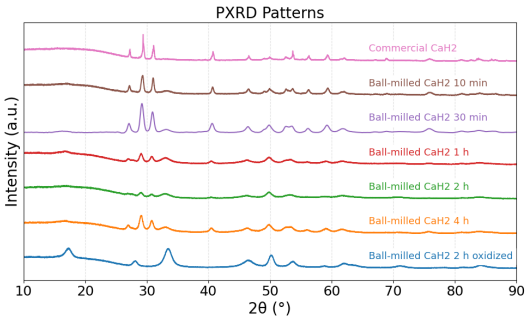
4 Results and Discussion

4.1 Morphology and Composition of Commercial and Ball-Milled CaH_2

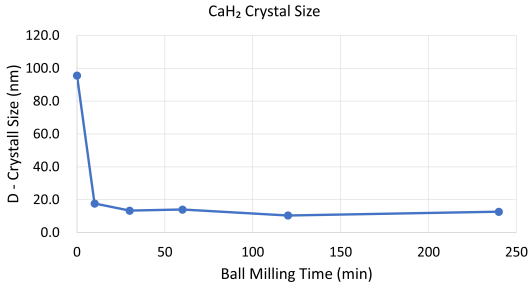
Figure 1a shows the XRD patterns of CaH_2 samples after various ball milling durations. Peak broadening was observed in all milled samples, increasing with milling time except at 4 h, where a slight reversal occurred. However, peak positions remained unchanged, indicating that no phase transformation occurred during milling. This consistent peak positioning confirms that no new crystalline phases formed. The reduced broadening at 4h may suggest partial agglomeration or recovery effects, which counteract further size reduction despite continued milling.

The average crystallite sizes were estimated using the Scherrer equation based on the peak near $2\theta \approx 29°$, as shown in Figure1b. The crystallite size of the 10 min milled CaH_2 sample was 17.6 nm, which is approximately $\frac{17.6}{95.5} \approx 18.4\%$ that of the commercial sample (95.5 nm). Samples milled for 0.5, 1, 2, and 4h had crystallite sizes of 13.3, 14.0, 10.4, and 12.7nm, respectively, showing that extended milling beyond 30 minutes does not significantly affect crystallite size. The saturation of crystallite size may be attributed to equilibrium between fracturing and cold welding during prolonged milling, although the exact mechanism remains uncertain.

Additionally, the XRD pattern of a 2h ball-milled CaH_2 sample exposed to air overnight showed missing peaks and the appearance of new peaks at different 2θ positions. This suggests oxidation and the possible formation of new surface compounds. SEM images



(a) *XRD patterns of CaH_2.*



(b) *Crystallite sizes of CaH_2.*

Figure 1: *Characterization of ball-milled CaH_2.*

(Figure 2) visually confirm that particle sizes decreased substantially after ball milling, especially in the early stages, further supporting the structural refinement inferred from XRD analysis.

4.2 H_2-Releasing Capacity of Commercial and Ball-Milled CaH_2

Samples subjected to varying ball-milling durations were analyzed using TG, DTA, and H_2-TPD to evaluate their hydrogen release behavior. Since CaH_2 releases only hydrogen upon heating, thermogravimetric anal-

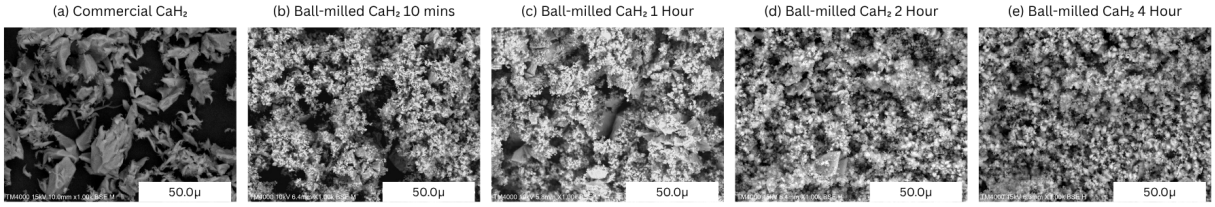


Figure 2: *SEM photos of Commercial, 10 mins, 1 hour, 2 hour, 4hour CaH_2*

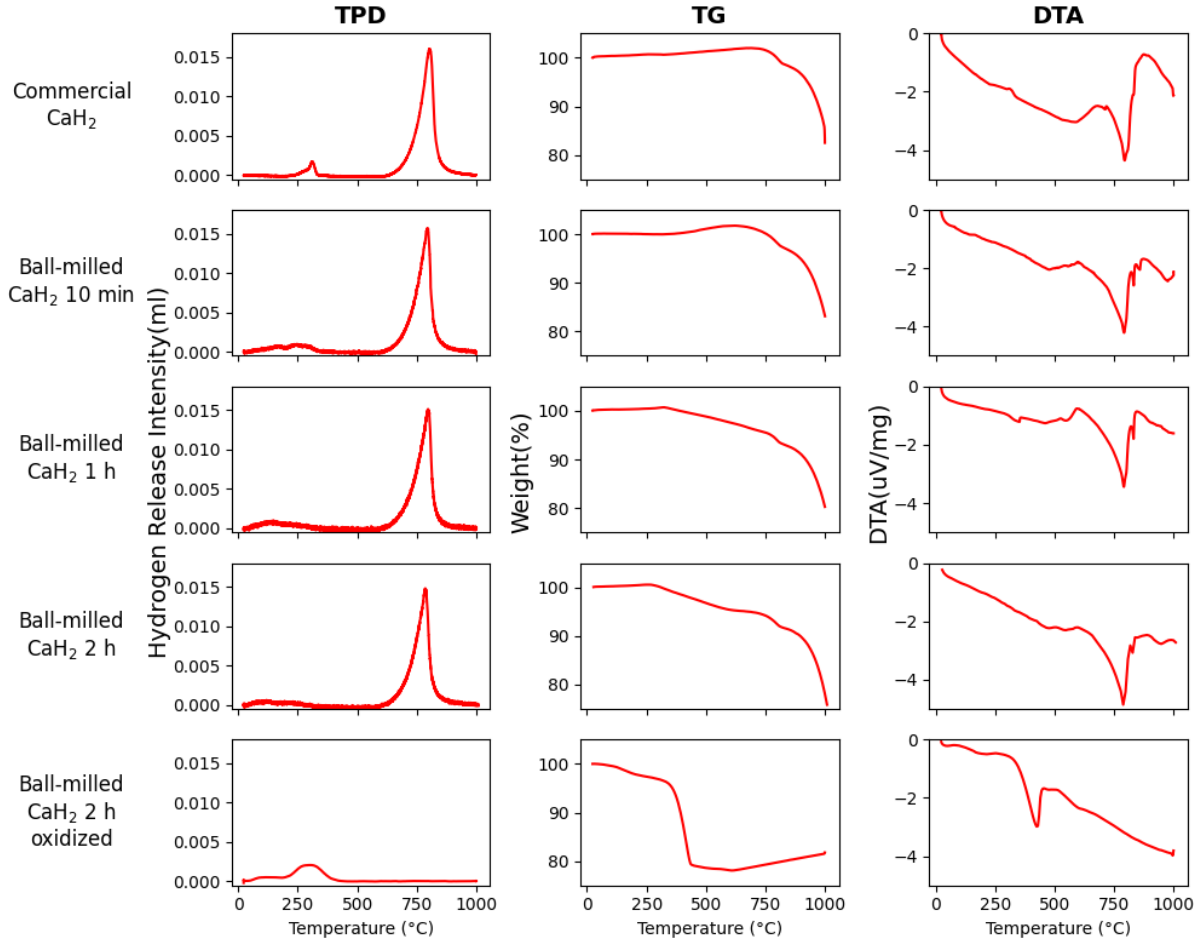


Figure 3: *TG, TPD, and DTA graphs of Commercial , 10 mins, 1 hour, 2 hour, 2hour oxidized CaH_2*

ysis followed by TPD with mass spectrometry allows for accurate quantification of H_2 release. Meanwhile, DTA provides insight into the heat of reaction 3. The TG and TPD curves for commercial CaH_2 3 shows two distinct hydrogen release peaks during heating to 1000°C, indicating a two-stage release mechanism. A relatively low-intensity first peak is followed by a more prominent second peak.

Ball milling significantly altered the first-stage H_2 release behavior. The onset temperature for this first peak shifted markedly from around 300°C in the commercial sample to approximately 50°C in ball-milled samples, with the peak also becoming noticeably broader. In contrast, the second peak exhibited only a minor decrease in release temperature—roughly 10°C—with little variation among the differently milled samples. This behavior is likely linked to the disruption of the surface oxide layer during ball milling, which increased exposure of the underlying CaH_2. Additionally, the reduced crystallite and particle sizes likely contributed to the altered hydrogen release kinetics.

We also performed XPS measurements on the series of CaH_2 samples. The Ca 2p peak profiles were found to be nearly identical across different milling durations, as well as in the air-exposed sample. These results suggest that the surface was already covered with an oxidized or hydroxidized layer, even with minimal air exposure, which may explain the limited effect of milling on the

second hydrogen release peak.

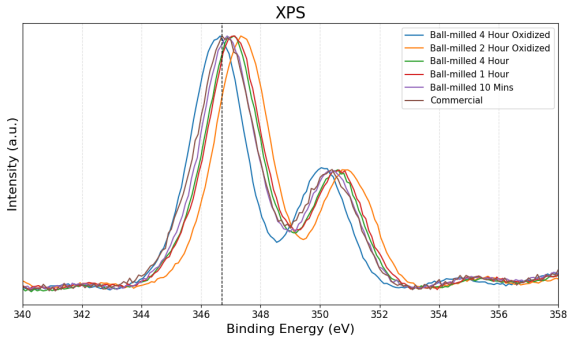


Figure 4: *XPS Scan of Ball-milled CaH_2*

4.3 Discussion

The observed improvements in the first hydrogen release peak suggest that ball milling effectively enhances the surface reactivity of CaH_2. This enhancement is likely due to increased defect density and reduced particle size, both of which improve hydrogen diffusion and activation at lower temperatures.

In contrast, the second peak remained largely unaffected by milling, indicating that bulk desorption properties are less sensitive to mechanical processing. One possible explanation is the presence of a residual surface oxide layer, which may have formed during sample transfer between instruments despite inert atmosphere precautions. Such a layer could act as a kinetic barrier, insulating the bulk material from the milling-induced modifications.

To investigate oxidation effects, a milled sample was intentionally exposed to air overnight. However, XRD analysis revealed that instead of forming CaO, the material had largely converted to calcium hydroxide ($Ca(OH)_2$), likely due to moisture exposure. This suggests that CaH_2 is highly reactive to ambient humidity and that air exposure leads to hydrolysis rather than pure oxidation. This also highlights a limitation in controlling surface chemistry post-milling and suggests that future efforts should focus on more robust handling techniques or in-situ characterization.

Further studies are needed to determine whether targeted surface treatments or controlled atmosphere annealing could suppress oxide formation and improve hydrogen release in the second stage

5 Conclusion

This study demonstrates that ball milling significantly enhances the first-stage hydrogen release properties of CaH_2, lowering the onset temperature and broadening the release profile due to increased surface reactivity and reduced crystallite size. However, the second-stage main release remains largely unchanged, likely due to persistent surface layers such as oxides or hydroxides. These findings highlight both the potential and limitations of mechanical processing in optimizing pure CaH_2 for hydrogen storage applications, emphasizing the need for improved surface control and handling techniques.

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